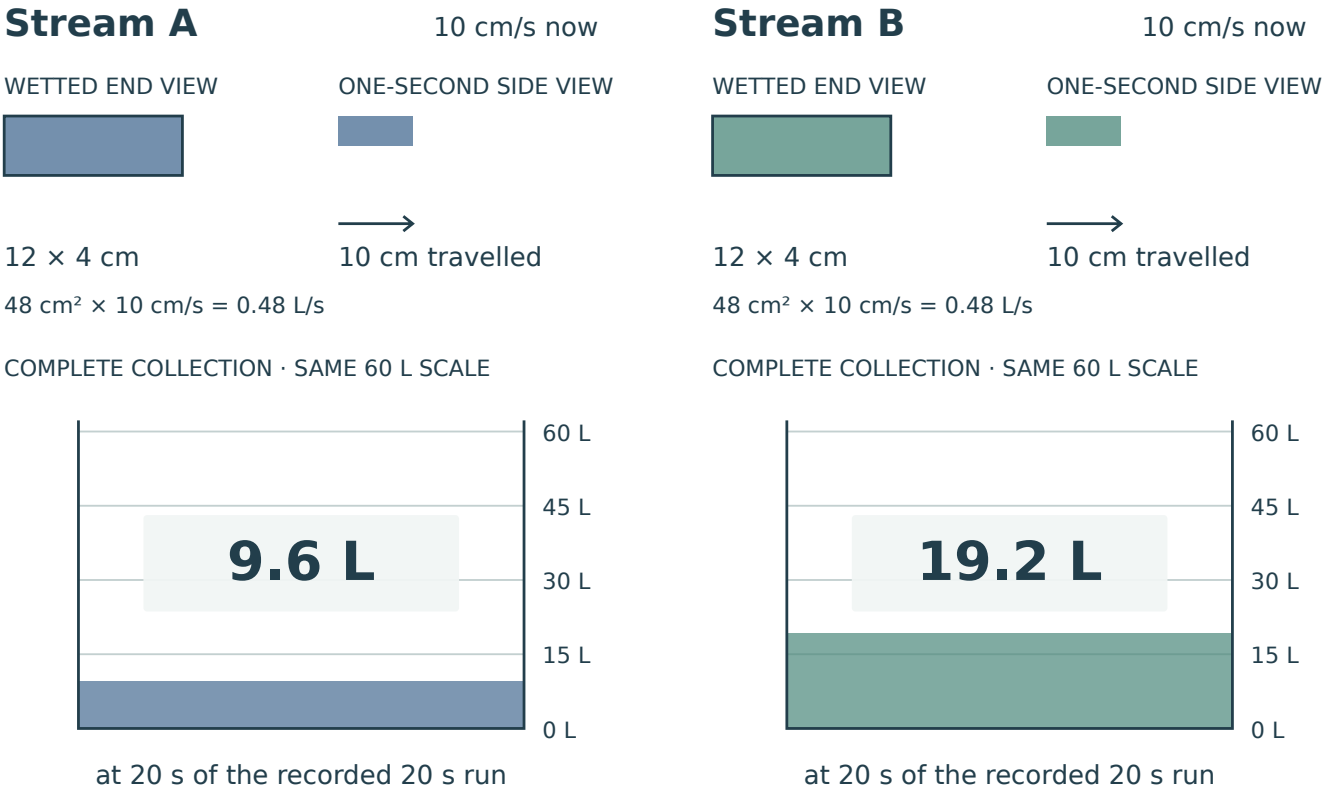
Water needs a clock.

Equal sections can deliver unequal supplies. A short observation can misrepresent a changing flow.

Reopen these settings. This file embeds its diagrams, full chapter reading, four printed pages, fonts and licenses. It makes no external subresource requests.

The collection experiment

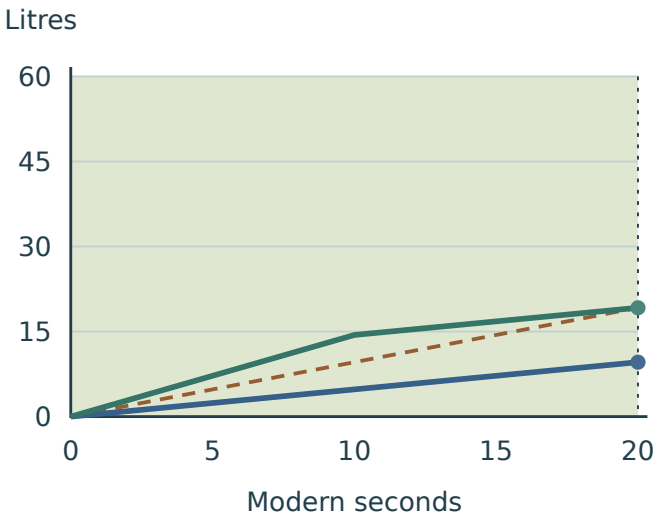
Both wetted sections are 12 × 4 cm, area 48 cm². A travels at 10 cm/s. B travels at 30 cm/s for the first ten seconds, then 10 cm/s. The saved frame is 20 s.



A modern ideal experiment: centimetres, seconds, litres, uniform section-mean speeds and the twenty-second schedule are chosen. All outflow is collected without leakage or storage delay. These are not measurements of an ancient spring or a conversion of Hero’s fingers and sundial hours.

For the uniform mean speeds used here, supply = wetted area × mean speed. A litre is 1,000 cm³. The modern equation explains the experiment; the chapter gives no algebraic discharge formula.

Measure an interval



Blue: A. Green: B. Dashed ochre: B’s selected average extended from an empty receiver to twenty seconds. Shading marks the measured window; the dotted vertical line marks the saved replay frame.

Selected interval: 0–20 s. Volume divided by elapsed time gives an interval average.

Stream	Start / L	End / L	Collected / L	Average / L/s
A	0	9.6	9.6	0.48
B	0	19.2	19.2	0.96

The selected B average is 19.2 L ÷ 20 s = 0.96 L/s. Applied to the whole twenty seconds, it predicts 19.2 L. The complete record contains 19.2 L. Difference: 0 L.

A short interval measures its own average. Extending it requires the flow to remain representative. The whole record’s average remains valid even when the instantaneous speed changes. At the exact ten-second switch, the displayed current speed is the speed after the change; cumulative volume remains continuous.

Five checkpoints from the full twenty-second record

Time / s	A / L	B / L
0	0	0
5	2.4	7.2
10	4.8	14.4
15	7.2	16.8
20	9.6	19.2

Read the complete chapter

Eleven units from Schöne’s edited chapter 31; Greek adapted from the Harvard College Library / Open Greek and Latin transcription, with a new English translation and language notes. Greek and new teaching text: CC BY-SA 4.0. The 1903 German translation is retained in the facing scans.

Dioptra 31 · p. 284, lines 11–12

1. Investigate a spring

Πηγῆς ὑπαρχούσης ἐπισκέψασθαι τὴν ἀπόρρυσιν αὐτῆς, τουτέστι τὴν ἀνάβλυσιν, ὅση ἐστίν.

Given a spring, to investigate its outflow—that is, how much it wells up.

πηγῆς ὑπαρχούσης · a spring being present

A genitive absolute supplies the starting condition. The aorist infinitive ἐπισκέψασθαι states the task: ‘to investigate’. It does not say the investigation took place in the past.

Dioptra 31 · p. 284, lines 12–13

2. The supply does not stay the same

εἰδέναι μέντοι χρὴ ὅτι οὐκ ἀεὶ ἡ ἀνάβλυσις ἡ αὐτὴ διαμένει.

One must, however, know that the flow does not always remain the same.

οὐκ ἀεὶ … ἡ αὐτή · not always … the same

Χρή with an infinitive expresses necessity: ‘one must know’. The article in ἡ αὐτή makes αὐτή mean ‘the same’. This opening matters when a short observation is extended to a longer period.

Dioptra 31 · p. 284, lines 14–18

3. Rain and drought

ὄμβρων μὲν γὰρ ὄντων ἐπιτείνεται διὰ τὸ ἐπὶ τῶν ὀρῶν τὸ ὕδωρ πλεονάζον βιαιότερον ἐκθλίβεσθαι, αὐχμῶν δὲ ὄντων ἀπολήγει ἡ ῥύσις διὰ τὸ μὴ ἐπιφέρεσθαι πλέον ὕδωρ. αἱ μέντοι γενναῖαι πηγαὶ οὐ παρὰ πολὺ τὴν ἀνάβλυσιν ἴσχουσιν.

For when there are rains it increases, because the abundant water on the mountains is forced out more strongly; but when there are droughts the flow falls away, because no more water is brought in. Strong springs, however, vary their flow only a little.

ὄμβρων μὲν … αὐχμῶν δέ · when there are rains … but when there are droughts

Two genitive absolutes balance rain against drought with μέν … δέ. Διὰ τὸ plus an infinitive gives a reason. The edition’s apparatus records corrections in the rain sentence; this reading follows the printed text. The experiment’s abrupt ten-second switch is a teaching device, not a model of rainfall response.

Dioptra 31 · p. 284, lines 18–21

4. Capture all the water

δεῖ οὖν περιλαβόντα τὸ πᾶν τῆς πηγῆς ὕδωρ, ὥστε μηδαμόθεν ἀπορρεῖν, σωλῆνα τετράγωνον μολιβοῦν ποιῆσαι, στοχασάμενον μᾶλλον μείζονα πολλῷ τῆς ἀποθύσεως·

One must therefore enclose all the spring's water, so that none flows away elsewhere, and make a square lead pipe, judging it to be considerably larger than the discharge requires.

τὸ πᾶν … ὕδωρ · all the water

Πᾶν agrees with the neuter ὕδωρ. The instruction is about complete capture: a receiver that misses some of the water understates the supply. The pipe is square; the water need not fill its opening. The model uses an ideal rectangular stream and complete collection.

Dioptra 31 · pp. 284, line 21 – 286, line 1

5. Use the dioptra to find the low outlet

εἶτα δι' ἑνὸς τόπου ἐναρμόσαι αὐτὸν ὥστε δι' αὐτοῦ τὸ ἐν τῇ πηγῇ ὕδωρ ἀπορρεῖν. δεῖ δὲ αὐτὸν κεῖσθαι εἰς τὸν ταπεινότερον τῆς πηγῆς τόπον, ὥστε ἔχειν αὐτὴν ἀπόρρυσιν· τὸν δὲ ταπεινότερον ἐπιγνωσόμεθα τῆς πηγῆς τόπον διὰ τῆς διόπτρας.

Then fit it through one place so that the water in the spring flows out through it. It must lie at the spring's lower place, so that the spring has an outlet; we shall identify the spring's lower place by means of the dioptra.

διὰ τῆς διόπτρας · by means of the dioptra

Διά with the genitive names the means. Ταπεινότερον is comparative: 'lower'. Here the surveying instrument chooses a place for the outlet; the later sundial supplies time. They do different jobs.

Dioptra 31 · p. 286, lines 1–6

6. Six times two makes twelve

ἀπολήψεται οὖν τὸ ἀπορέον διὰ τοῦ σωλῆνος ὕδωρ ἐν τῷ περιστομίῳ τοῦ σωλῆνος· οἷον ἀπολαμβάνει δακτύλους β· ἐχέτω δὲ καὶ τὸ πλάτος τοῦ περιστομίου τοῦ σωλῆνος δακτύλους ς· ἑξάκις δύο γίνονται ιβ· ⟨ἀποφανούμεθα δὴ τὴν ἀνάβλυσιν τῆς πηγῆς δακτύλων ιβ⟩.

The water flowing through the pipe will occupy part of its mouth: for example, it occupies two finger-widths in depth; let the mouth's width also be six finger-widths. Six times two makes twelve. ⟨We shall then declare the spring's flow to be twelve fingers.⟩

ἑξάκις δύο · six times two

B = 2, ς = 6, ιβ = 12. 'In depth' makes explicit the dimension paired with width, as Schöne's German does. The product measures a cross-sectional area even though the text says 'fingers'. It is not a volume per time. Angle brackets preserve the editor's relocated conclusion; the final duplicate is deleted in the printed edition.

Dioptra 31 · p. 286, lines 6–9

7. Size is insufficient

εἰδέναι δὲ χρὴ ὅτι οὐκ ἔστιν αὔταρκες πρὸς τὸ ἐπιγνῶναι, πόσον χορηγεῖ ὕδωρ ἡ πηγή, τὸ εὑρεῖν τὸν ὄγκον τοῦ ῥεύματος, ὃν λέγομεν εἶναι δακτύλων ιβ,

But one must know that finding the size of the stream, which we call twelve fingers, is not sufficient to discover how much water the spring supplies;

οὐκ ἔστιν αὔταρκες · it is not sufficient

The subject is the articular infinitive τὸ εὑρεῖν, 'finding'. Ὄγκος is rendered 'size' here, not automatically modern three-dimensional 'volume': the preceding two lengths produced an area. This sentence continues into the next unit.

Dioptra 31 · p. 286, lines 9–11

8. The missing quantity is speed

ἀλλὰ καὶ τὸ τάχος αὐτοῦ· ταχυτέρας μὲν γὰρ οὔσης τῆς ῥύσεως πλέον ἐπιχορηγεῖ τὸ ὕδωρ, βραδυτέρας δὲ μεῖον.

its speed is needed too. For when the flow is faster it supplies more water; when slower, less.

τάχος / πάχος · speed / thickness

The edited text reads τάχος, ‘speed’; Schöne’s apparatus at lines 9–10 records τὸ πάχος, ‘thickness’. A single letter changes the quantity. This lesson follows the edited reading and the following faster/slower comparison; it does not present that choice as an autograph we can inspect. Ταχυτέρας and βραδυτέρας are feminine comparatives agreeing with ῥύσεως.

Dioptra 31 · p. 286, lines 11–14

9. Collect the water and observe the time

διὸ δεῖ ὑπὸ τὴν τῆς πηγῆς ῥύσιν ὀρύξαντα τάφρον τηρῆσαι ἐξ ἡλιακοῦ ὡροσκοπίου, ἐν τινὶ ὥρᾳ πόσον ἀπορρεῖ ὕδωρ ἐν τῇ τάφρῳ,

Therefore one must dig a trench beneath the spring’s outflow and observe by a sundial how much water flows into the trench in a certain interval;

ἐξ ἡλιακοῦ ὡροσκοπίου · by a solar time indicator

Ἡλιακοῦ describes the instrument as solar: a sundial. Τάφρος is a trench or ditch used here to receive the water. The text supplies no numerical timing interval or receiver dimensions. Our graduated vessels and modern seconds make the method visible; they are not a reconstruction of that equipment.

Dioptra 31 · p. 286, lines 14–15

10. Estimate a longer supply

καὶ οὕτως στοχάσασθαι τὸ ἐπιχορηγούμενον ὕδωρ ἐν τῇ ἡμέρᾳ πόσον ἐστίν,

and in this way estimate how much water is supplied in the day,

στοχάσασθαι · to estimate

An aorist middle infinitive continues δεῖ, ‘one must’. The wording calls for an estimate. Applying a short sample’s rate to a longer period requires the sample to represent that period. The changing-flow exercise exposes that assumption; it does not convert an ancient hour into a fixed number of seconds.

Dioptra 31 · p. 286, lines 15–17

11. The opening need not be measured

ὥστ’ οὐδὲ ἀναγκαῖόν ἐστι τὸν ὄγκον τῆς ῥύσεως τηρεῖν· διὰ γὰρ τοῦ χρόνου δήλη ἐστὶν ἡ χορηγία.

so that it is not even necessary to observe the size of the flowing stream: through the time, the supply becomes clear.

διὰ τοῦ χρόνου · through the time

The article and genitive follow διά. The conclusion depends on the preceding collection of water as well as the time observation. Time alone cannot determine a volume. Measuring collected volume over an interval bypasses a separate measurement of the opening and speed.

What the evidence does and does not say

The source example multiplies two finger-widths by six. It then calls the result twelve fingers. The product is a section, not a supply per time. No fixed finger-to-centimetre conversion is used here. The source names a sundial and an interval, without a numerical duration; our modern seconds do not reconstruct its hours.

The printed text reads τάχος, speed, while the apparatus records τὸ πάχος, thickness, at p. 286, lines 9–10. The model follows the edited text and its faster/slower comparison. Schöne relocates the twelve-finger conclusion into angle brackets and deletes its duplicate at the end. These are editorial decisions, not invisible repairs.

Line-end hyphens and whitespace are joined; apparatus, line numbers and deleted letters are omitted in the reading. The digital duplicate “μενον” after στοχασάμενον is removed following the page; ὀρῶν follows the printed mountain word. The full scans retain all editorial marks. The language notes distinguish interpretation from literal wording.

Collection is exact and complete in the model. Receiver shape, scale and centimetre dimensions are teaching choices. There is no pressure, friction, rainfall, leakage or instrument-error simulation. The chapter’s rain discussion motivates asking whether supply changes; it supplies no ten-second switching schedule.

284 · Greek: the changing spring

Chapter 31 begins at line 11 below the end of chapter 30. The lower apparatus records corrections in the rain passage. The pipe instruction continues on page 286. Source volume.

τῆς περιμέτρου τῆς $B\Gamma$· ⟨ἡ δὲ BE, ἥ ὑπερέχει ἡ ἡμίσεια τῆς περιμέτρου τῆς $A\Gamma$⟩, ἡ δὲ ΓE, ἥ ὑπερέχει ἡ ἡμίσεια τῆς περιμέτρου τῆς AB· δοθὲν ἄρα καὶ τὸ ἐμβαδὸν ⟨τοῦ⟩ τριγώνου. συντεθήσεται δὴ οὕτως· ἔστω ἡ μὲν AB μοιρῶν ιγ, ἡ δὲ $B\Gamma$ μοιρῶν ιδ, ἡ δὲ ΓA μοιρῶν ιε. σύνθες τὰς τρεῖς, γίνονται μβ· τούτων τὸ ἥμισυ κα. ἄφελε τὰ ιγ, λοιπὸν η· καὶ τὰ ιδ, λοιπὸν ζ· καὶ τὰ ιε, λοιπὸν ϛ. τὰ κα, η, ζ, ϛ ⟨πολλαπλασιασθέντα⟩ δι' ἀλλήλων γίνονται ,ζνϛ· τούτων ἡ πλευρὰ ἔσται πδ. τὸ ἐμβαδὸν τοῦ τριγώνου πδ.

λα. Πηγῆς ὑπαρχούσης ἐπισκέψασθαι τὴν ἀπόρρυσιν αὐτῆς, τουτέστι τὴν ἀνάβλυσιν, ὅση ἐστίν. εἰδέναι μέντοι χρὴ ὅτι οὐκ ἀεὶ ἡ ἀνάβλυσις ἡ αὐτὴ διαμένει. ὄμβρων μὲν γὰρ ὄντων ἐπιτείνεται διὰ τὸ ἐπὶ τῶν ὀρῶν τὸ ὕδωρ πλεονάζον βιαιότερον ἐκθλίβεσθαι, αὐχμῶν δὲ ὄντων ἀπολήγει ἡ φύσις διὰ τὸ μὴ ἐπιφέρεσθαι πλέον ὕδωρ. αἱ μέντοι γενναῖαι πηγαὶ οὐ παρὰ πολὺ τὴν ἀνάβλυσιν ἴσχουσιν. δεῖ οὖν περιλαβόντα τὸ πᾶν τῆς πηγῆς ὕδωρ, ὥστε μηδαμόθεν ἀπορρεῖν, σωλῆνα τετράγωνον μολιβοῦν ποιῆσαι, στοχασάμενον μᾶλλον μείζονα πολλῷ τῆς ἀποθύσεως· εἶτα δι' ἑνὸς τόπου ἐναρμόσαι αὐτὸν ὥστε δι' αὐτοῦ τὸ ἐν τῇ πηγῇ ὕδωρ ἀπορρεῖν. δεῖ δὲ αὐτὸν κεῖσθαι εἰς τὸν ταπεινότερον τῆς πηγῆς τόπον, ὥστε ἔχειν αὐτὴν ἀπόρρυσιν· τὸν δὲ ταπεινότερον ἐπιγνωσόμεθα τῆς πηγῆς

p. 290 p. 294

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6 *συνθέντας τὰς*: correxi 9 *ΖΗϛ* 10 *ΗΔ το* 14—15 *ἐπιτίθεται διατίθεται δια τὸ επὶ τῶν ὡρῶν*: correxi coll. Anonymo Byz. p. 390, 1 Vi 15 *πλεονάζειν βιαιότερον ἐκθλιβόμενον*: correxi coll. anonymo Byz. p. 390, 2 Vinc. Similes corruptelae apud Philonem. Mech. Synt. l. V p. 80, 14 a C. Graux et apud Dionysium de imitatione p. 20, 21 ab H. Usenero sublatae sunt 17 *γένναι αἱ* 20 *μολιβον* 24 *αὐτὸν*: correxi

285 · German: the opening instruction

Schöne's facing German begins chapter XXXI near the bottom. The numerical triangle example above belongs to chapter 30. Source volume.

Mithin: $\Gamma B : B\Delta = \Delta\Delta : \Delta H = \Theta B : HE$ und
$\Gamma B : B\Theta = B\Delta : HE = BK : KE$ und
$\Gamma\Theta : \Theta B = BE : EK$. Daher auch $\Gamma\Theta^2 : \Gamma\Theta \times \Theta B = BE \times E\Gamma : \Gamma E \times EK = BE \times E\Gamma : HE^2$.

Daher wird das Produkt aus dem Quadrat von $\Gamma\Theta$ und dem Quadrat von EH, aus dem die Wurzel gleich dem Dreieck war, gleich $\Gamma\Theta \times \Theta B \times \Gamma E \times EB$ sein. Und jede der Geraden $\Gamma\Theta$, ΘB, BE und $E\Gamma$ wird gegeben sein. Denn $\Gamma\Theta$ ist gleich der Hälfte des Umfangs, ΘB gleich der Differenz des halben Umfangs und der Geraden $B\Gamma$; BE ist gleich der Differenz des halben Umfangs und der Geraden $A\Gamma$; ΓE ist gleich der Differenz des halben Umfangs und der Geraden AB. Also ist auch der Inhalt des Dreiecks gegeben.

Berechnet wird es folgendermafsen. Es sei $AB = 13$, $B\Gamma = 14$, $\Gamma A = 15$.

$$13 + 14 + 15 = 42$$
$$\frac{42}{2} = 21$$
$$21 - 13 = 8$$
$$21 - 14 = 7$$
$$21 - 15 = 6$$
$$21 \times 8 \times 7 \times 6 = 7056$$
$$\sqrt{7056} = 84.$$

der Inhalt des Dreiecks ist $= 84$.

XXXI. Wenn eine Quelle vorhanden ist, ihren Abfluſs, d. h. die Menge des Wassers, das sie aufsprudeln läſst, zu untersuchen.

Man muſs jedoch wissen, daſs der Ausfluſs sich nicht stets gleich bleibt. Denn wenn Regenzeit ist, so wird er stärker, weil dann das Wasser auf den Bergen in gröſseren Mengen vorhanden ist und mit stärkerer Gewalt aus dem Boden herausgepreſst wird; herrscht dagegen Trockenheit, so hört der Abfluſs auf, weil nicht mehr Wasser zuströmt.

SHA-256 14dfed8a60a62893904677ff16df23e3793c09b5cf7162391b57e5b1f4def638

286 · Greek: size, speed and time

The chapter concludes at line 18. Notice the supplied twelve-finger conclusion near line 5 and its deleted duplicate at the end. The apparatus at lines 9–10 records τὸ πάχος where the edited text reads τὸ τάχος. Source volume.

τόπον διὰ τῆς διόπτρας. ἀπολήψεται οὖν τὸ ἀπορρέον διὰ τοῦ σωλῆνος ὕδωρ ἐν τῷ περιστομίῳ τοῦ σωλῆνος· οἷον ἀπολαμβάνει[ν] δακτύλους β· ἐχέτω δὲ καὶ τὸ πλάτος τοῦ περιστομίου τοῦ σωλῆνος δακτύλους ς· ἑξάκις δύο γίνονται ιβ· ⟨ἀποφανούμεθα δὴ τὴν ἀνάβλυσιν τῆς πηγῆς δακτύλων ιβ⟩. εἰδέναι δὲ χρὴ
fol. 76ᵛ ὅτι οὐκ ἔστιν αὔ|ταρκες πρὸς τὸ ἐπιγνῶναι, πόσον χορηγεῖ ὕδωρ ἡ πηγή, [ἢ] τὸ εὑρεῖν τὸν ὄγκον τοῦ ῥεύματος, ὃν λέγομεν εἶναι δακτύλων ιβ, ἀλλὰ καὶ τὸ
p. 296 τάχος αὐτοῦ· ταχυτέρας μὲν γὰρ οὔσης τῆς ῥύσεως πλέον ἐπιχορηγεῖ τὸ ὕδωρ, βραδυτέρας δὲ μεῖον. διὸ δεῖ ὑπὸ τὴν τῆς πηγῆς ῥύσιν ὀρύξαντα τάφρον τηρῆσαι ἐξ ἡλιακοῦ ὡροσκοπίου, ἐν τινὶ ὥρᾳ πόσον ἀπορρεῖ ὕδωρ ἐν τῇ τάφρῳ, καὶ οὕτως στοχάσασθαι τὸ ἐπιχορηγούμενον ὕδωρ ἐν τῇ ἡμέρᾳ πόσον ἐστίν, ὥστ᾽ οὐδὲ ἀναγκαῖόν ἐστι τὸν ὄγκον τῆς ῥύσεως τηρεῖν· διὰ γὰρ τοῦ χρόνου δήλη ἐστὶν ἡ χορηγία. [ἀποφανούμεθα δὴ τὴν ἀνάβλυσιν τῆς πηγῆς δακτύλων ιβ].

λβ. Ἐπεὶ οὖν διὰ τῆς κατασκευασθείσης ἡμῖν διόπτρας τὰς ἐπὶ γῆς χρείας πρὸς τὰς διοπτρικὰς ἐπαγγελίας ἀπεδείξαμεν, εὔχρηστον δέ ἐστιν εἰς πολλὰ καὶ πρὸς τὰ οὐράνια πρὸς τὸ τὰς τῶν ἀπλανῶν ἀστέρων ἢ παὶ τῶν πλανητῶν ἀποστάσεις εἰδέναι, ἀποδείξομεν διὰ τῆς διόπτρας ὡς δεῖ καὶ τὰ ⟨τούτων⟩ ἀποστήματα λαμβάνειν. ἐν γὰρ τῷ ὑπὸ γαστέρα τοῦ τυμπάνου τοῦ ἐν τῇ διόπτρᾳ κύκλον γράψομεν περὶ τὸ αὐτὸ κέντρον

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3 ἀπολαμβάνειν: correxi 4 τὸ περιστόμιον 8 πηγὴ ἢ τὸ εὑρεῖν 9—10 τὸ πάχος 11 f. ἐπιχορηγεῖται ὕδωρ 11—12 διο δη 17—18 δὲ τὴν 18 δακτύλων δεδεκα (sic); haec transposui in vs. 5. 19 διὰ deleverim 24 ⟨τούτων⟩ addidi 26 τω αυτω κεντρω, sed ex τῷ αὐτῷ fec. τὸ αὐτὸ man. 1

SHA-256 10871f5164d030766903fa41cd037a784cfa1b7bf18a6d8d242782b4d0d2e36f

287 · German: the complete measurement

The facing German continues the enclosure and pipe instructions, then explains speed and timed collection. Chapter XXXII starts below the conclusion. Source volume.

Die guten Quellen reduzieren jedoch ihren Abfluſs nur um ein Geringes. Man muſs nun die ganze Wasserfläche der Quelle einfassen, so daſs nirgends etwas abflieſsen kann und dann eine Bleiröhre von quadratischem Querschnitt herstellen, indem man darauf sieht, daſs dieselbe um ein Bedeutendes gröſser ist als der regelmäſsige Abfluſs verlangt. Sodann muſs man diese an einer Stelle so einsetzen (in die Umfassungsmauer), daſs das Quellwasser durch dieselbe abflieſst. Diese Stelle muſs nach der Stelle zu liegen, die niedriger als die Quelle liegt, so daſs sie Abfluſs hat. Die Stelle aber, welche tiefer als die Quelle liegt, werden wir vermittelst der Dioptra ermitteln. Das durch die Röhre abflieſsende Wasser wird nun an der Öffnung der Röhre einen gewissen Raum einnehmen. Beispielsweise nimmt es 2 Daktylen (in der Höhe) ein, die Breite aber der Öffnung der Röhre soll 6 Daktylen betragen. $6 \times 2 = 12$; wir werden daher den Abfluſs der Quelle auf 12 Daktylen angeben. Man muſs jedoch wissen, daſs es, um zu erkennen, wie viel Wasser die Quelle liefert, nicht genügt, die Ausdehnung des Abfluſsstroms zu kennen, welche nach unserer Behauptung 12 Daktylen beträgt, sondern man auch seine Geschwindigkeit kennen muſs. Denn ist der Abfluſs ein geschwinderer, so liefert die Quelle mehr, ist er ein langsamerer, so liefert sie weniger Wasser.

Man muſs daher unterhalb des Quellabflusses ein Reservoir graben und mit einer Sonnenuhr beobachten, welches Quantum Wassers in einer bestimmten Zeit abflieſst und so annähernd bestimmen, wie groſs die Quantität des an einem Tage gelieferten Wassers ist. Es ist daher (bei dieser Methode) gar nicht einmal nötig, die Gröſse des Abfluſsstromes zu beobachten, denn die Leistungsfähigkeit wird durch die Zeit klar.

XXXII. Da wir nun vermittelst der von uns konstruierten Dioptra die Verwendung des Instrumentes auf der Erdoberfläche bei dioptrischen Problemen nachgewiesen haben, dieselbe jedoch nach vielen Richtungen auch für

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